SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA5101
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

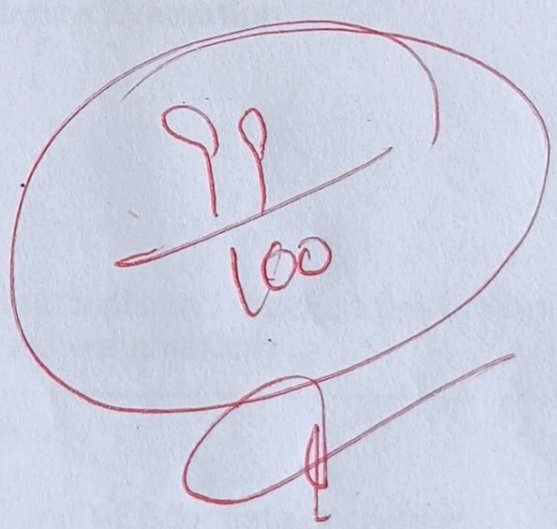
QUESTIONS: **Total Marks: 50**

Annexure A

Code of Conduct:

I Shaik Asha (192473025) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



Annexure A
Analytical Problem Solving

1. RSA Key Generation and Security Analysis
Given:
$p = 17$, $q = 19$
Compute n and $\phi(n)$
Choose $e = 5$ and calculate the private key d
Encrypt the message $M = 7$
Decrypt the ciphertext and verify correctness
Analyze the impact of choosing a small prime number on RSA security

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2. ElGamal Encryption Analysis
Given:
$p = 23$, $g = 5$, private key $x = 7$, random $k = 3$, message $M = 10$
Compute the public key
Perform encryption to generate ciphertext pair (C1, C2)
Decrypt the ciphertext to retrieve the original message
Analyze how randomness (k) improves security in ElGamal

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3. RSA Digital Signature Verification
Given:
$p = 11$, $q = 17$, $e = 7$, message $M = 8$
Compute n and $\phi(n)$
Find private key d
Generate the digital signature for the message
Verify the signature using the public key
Analyze how digital signatures ensure non-repudiation

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4. Diffie-Hellman with Attack Resistance Evaluation
Given:
$p = 29$, $g = 2$
User A private key = 5
User B private key = 12
Compute public keys for both users
Compute the shared secret key
Analyze what happens if an attacker intercepts and replaces public keys
Suggest a cryptographic solution to ensure authenticity

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5. Hybrid Encryption System Analysis
Given:
A system uses RSA for key exchange and AES for data encryption
RSA encryption time = 5 ms per key exchange
AES encryption time = 0.5 ms per data block
Total data blocks = 500
Calculate total time taken for encryption
Compare with using only RSA for encrypting all data

Analyze why hybrid encryption is more efficient
Justify its use in modern secure communication systems

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20% 20	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20% 20	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25% 24	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process

Accuracy of Calculation	20% 20	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15% 15	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

95

RSA Key Generation and Security Analysis

Given Data: Prime number, $p = 17$

Prime number, $q = 19$

Public exponent, $e = 5$

Message, $M = 7$

To find: n, $\phi(n)$, private key d, ciphertext C, decrypted message, and the impact of small prime numbers on RSA security.

Step-1: Calculate n

RSA modulus is calculated using:

$$n = p \times q$$

$$= 17 \times 19$$

$$\boxed{n = 323}$$

Step-2: Calculate Euler's Totient Function

Since p and q are distinct prime numbers:

$$\phi(n) = (p-1)(q-1)$$

$$\phi(n) = (17-1)(19-1) = 16 \times 18$$

$$\boxed{\phi(n) = 288}$$

Step-3: Calculate the Private Key d

The private exponent d is the modular multiplicative inverse of e modulo $\phi(n)$.

$$d \times e \equiv 1 \pmod{\phi(n)}$$

Substituting $e = 5$:

$$5d \equiv 1 \pmod{288}$$

Therefore, we need an integer d such that $5d$ leaves a remainder of 1 when divided by 288.

Choose $d = 173$,

$$5 \times 173 = 865$$

$$865 = 3 \times 288 + 1$$

Hence, $865 \% 288 = 1$

$$\therefore d = 173$$

The public key is $(e, n) = (5, 323)$, and the private key is $(d, n) = (173, 323)$

Step-4: Encrypt the Message

RSA encryption formula; $C = M^e \bmod n$

Given, $M = 7, e = 5, n = 323$

$$C = 7^5 \bmod 323$$

$$C = 16807 \bmod 323 = 11$$

$$\therefore C = 11$$

The ciphertext is 11.

Step-5: Decrypt the Ciphertext

RSA decryption formula:

$$M = C^d \bmod n$$

$$M = 11^{173} \bmod 323$$

$$\boxed{M = 7}$$

Step-6: Verify Correctness

Original Message: M = 7

Decrypted Message: M = 7

Since both values are equal, the encryption and decryption process is correct.

Step-7: Security Analysis of Small Prime Numbers

RSA security depends on the difficulty of factoring the modulus n into its prime factors p and q.

In this problem:

$$n = 323 = 17 \times 19$$

These prime factors are very small and can be discovered easily. Once an attacker finds p and q, the attacker can calculate $\phi(n)$ and determine the private exponent d.

The consequences of choosing small primes are:

1. Easy factorization: Attackers can quickly factor the modulus.
2. Private-key recovery: Knowing the prime factors allows an attacker to calculate the private key.
3. Reduced Confidentiality: Encrypted messages can be decrypted by an attacker who recovers the private key.

4. Insufficient Computational Security: Small moduli cannot withstand modern factoring techniques.

In practical applications, RSA requires a sufficiently large modulus, commonly 2048 bits or more, along with secure key generation and appropriate padding schemes. The small values in this exercise are intended only for learning.

Result: The RSA public key is (5, 323), the private key is (173, 323), and the ciphertext for message 7 is 11. Decryption successfully recovers the original message.

2. ElGamal Encryption Analysis

Given Data: Prime number $p = 23$

Generator $g = 5$

Private key $x = 7$

Random number $K = 3$

Message $M = 10$

To find: Public Key, ciphertext pair (C_1, C_2), decrypted message, and the role of randomness in ElGmal encryption.

Step-1: Calculate the Public Key

In ElGamal encryption, the public key component y is calculated using:

$$y = g^x \bmod p$$

Substituting the values:

$$y = 5^7 \bmod 23$$

Calculate powers of 5 modulo 23:

$$5^2 = 25 \equiv 2 \pmod{23}$$

$$5^4 \equiv 2^2 = 4 \pmod{23}$$

Since $7 = 4+2+1$:

$$5^7 \equiv 5^4 \times 5^2 \times 5 \pmod{23}$$

$$y \equiv 4 \times 2 \times 5$$

$$y = 40 \bmod 23$$

$$y = 17$$

Therefore:

Private Key, $x = 7$

Public Key, $(P, g, y) = (23, 5, 17)$

Step-2: Calculate the First Ciphertext Component C_1

ElGamal encryption uses the random number K to calculate:

$$C_1 = g^K \bmod p$$

Substituting the values:

$$C_1 = 5^3 \bmod 23$$

$$C_1 = 125 \bmod 23 \quad [125 = 23 \times 5 + 10]$$

$$C_1 = 10$$

Step-3: Calculate the second Ciphertext Component C_2

The second ciphertext component is calculated using:

$$C_2 = M \times y^K \bmod p$$

Substituting the values:

$$C_2 = 10 \times 17^3 \bmod 23$$

Now, $17^2 = 289 \equiv 13 \pmod{23}$

$$17^3 \equiv 13 \times 17 = 221 \pmod{23}$$

$$17^3 \equiv 14 \pmod{23}$$

Therefore,

$$C_2 = 10 \times 14 \bmod 23$$

$$C_2 = 140 \bmod 23$$

Since, $140 = 23 \times 6 + 2$

$$C_2 = 2$$

Hence, the ciphertext pair is:

$$(C_1, C_2) = (10, 2)$$

Step-4: Decrypt the Ciphertext

The receiver uses the private key x to calculate the shared secret:

$$S = C_1^x \bmod p$$

substituting the values,

$$S = 10^7 \bmod 23$$

Using modular exponentiation:

$$10^2 \equiv 8 \pmod{23}$$

$$10^4 \equiv 8^2 = 64 \equiv 18 \pmod{23}$$

Since, $7 = 4+2+1$

$$10^7 \equiv 18 \times 8 \times 10 \pmod{23}$$

$$10^7 \equiv 1440 \bmod 23$$

$$5 = 14$$

Next, calculate the modular inverse of 14 modulo 23.

We need a value 5^{-1} satisfying:

$$14\,5^{-1} \equiv 1 \pmod{23}$$

Since :

$$14 \times 5 = 70 = 23 \times 3 + 1$$

$$\therefore 5^{-1} = 5$$

The original message is recovered using:

$$M = C_2 \times 5^{-1} \bmod p$$

$$M = 2 \times 5 \bmod 23$$

$$M = 10$$

Step-5: Verify Correctness

Original message: $M = 10$

Decrypted message: $M = 10$

Both values match, confirming that the ElGamal encryption and decryption process is correct.

Step-6: Analyze How Randomness Improves Security in ElGamal

Randomness is an essential security feature of ElGamal encryption. The random number K is generated for each encryption operation.

Its importance can be explained as follows:

1. Different ciphertexts: Encrypting the same message multiple times with different random values of K normally produces different ciphertext pairs.

2. Protection against pattern analysis: Randomness makes it more difficult for pso attackers to identify repeated messages by comparing ciphertexts.

3. Improved confidentiality: An attacker cannot easily determine the original message merely by observing repeated encryptions of the same message.

However, the random number K must be generated securely and must not be reused carelessly. Reusing K can expose relationships between encrypted messages and may compromise confidentiality. The prime number and generator must also be selected appropriately.

Result: The ElGamal public key is (23, 5, 17), and the ciphertext pair is (10, 2). Decryption successfully recovers the original message.